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Description - Contains theory overview for the Full-Size PDU. Sheet numbers listed at the end of some subsection titles refer to the sheet number within the PDU block diagrams.

1- INTRODUCTION

The Power Distribution Unit (PDU) serves as the primary interface to facility power. It converts the 3-phase input line voltage to either 208Vac (3-phase) or 120Vac (1-phase) as required by each MR subsystem. In addition, the PDU provides protection against high voltage spikes caused by lightning or switching transients, and both transverse mode and common mode high frequency noise. No power line distortion or frequency shift protection is provided. Only the PDU with the integrated PowerTECH regulator (46-282368G4) provides active power line regulation and protection against power line surges and sags.

Temperature fault detection, and power distribution control and sequencing are also provided by the PDU. Status signals are sent from the PDU to the Signal Collecting Board (SCB) located in the system cabinet (MR2).

For nonregulated PDUs, refer to block diagram sheets 1 and 2. For PDU with PowerTECH, refer to block diagram sheets 3 through 5.

2- MAIN POWER

2-1 Facility Disconnect and Emergency Off

2-1-1 Nonregulated PDU - Sheet 1

Input power is first routed to the facility disconnect. The disconnect uses a low voltage, low energy (LVLE) transformer to provide power to an under voltage release device. This device is in series with the Emergency Off push buttons and the normally closed contact of K42 on the PDU control board. (Note that a contact is provided for a normally open configuration of the facility disconnect; however, this configuration is not recommended.)

The coil of K42 becomes energized when the transformer temperature exceeds 200°C. This in turn opens the circuit, causing the under voltage device to automatically trip the main breaker when its applied coil voltage drops to a specific percent of the rated value. The same effect can be achieved by actuating any of the Emergency Off push buttons.

2-1-2 PDU with PowerTECH Regulator - Sheet 3

Emergency off is accomplished by actuating any of the customer-supplied Emergency Off push buttons. A temperature monitor circuit within the regulator shunt trips the input breaker (CB1) if the regulator transformer temperature exceeds 160°C. This shuts down the incoming power to the PDU, but does not trip the facility disconnect.

2-2 Power Distribution

2-2-1 Nonregulated PDU - Sheet 1

The PDU accepts a 3-phase (delta or wye) 318-533 Vac (also 190-225 Vac for PDU which has a 75 kVA transformer), 47-63 Hz service from the facility input line. The maximum input power is 112 kVA, or 75 kVA, depending on which PDU transformer you have. MOV (metal oxide varistor) transient power protection is provided for each phase. The MOVs are in series with 30A/700V fuses. Neon indicators are in parallel with the MOVs, and are on unless the facility main breaker is open, or one or more MOV fuses are blown. These indicators operate as phase detectors. However, if a facility input phase is lost, current flow from the remaining phases through the transformer windings will cause the corresponding phase indicator to remain illuminated with reduced intensity, rather than going out.

Power transformer input tap selection is used to obtain the correct output voltages required by the subsystems (120 Vac 1-phase and 208 Vac 3-phase wye). The 3-phase output then passes through a 400A/240V circuit breaker (CB18) where it is distributed within the 3-phase load center. Note that the neutral and ground buses are tied together.

The load center contains all feeder circuit breakers. From the load center subsystem, power is distributed through contactors to each subsystem. Feeder 17 (PDU Power) and Feeder 16 (Service Receptacles) are the only feeders that are not contactor-controlled.

2-2-2 PDU with PowerTECH Regulator - Sheet 3

The PDU with PowerTECH is similar to the nonregulated PDU, the major difference being that the transformer of the nonregulated PDU is replaced with the PowerTECH regulator. The regulator is a 3-phase tapped voltage regulator. It consists of a multitapped transformer with logic control and SCR circuitry. The output voltage is regulated by electronically selecting an appropriate transformer tap. There are six transformer taps per phase. The taps are spaced at 3% intervals of the nominal input voltage. Tap 5 is located at the nominal input voltage. For an input voltage range of +4.5% surge and -13.5% sag, the regulator maintains the output voltage to within 1.5%.

The PDU with PowerTECH accepts a 3-phase (delta or wye) 480 Vac, 60 $\pm$ 0.5Hz, 75 kVA max. service from the facility input line. MOV (metal oxide varistor) transient power protection is provided for each phase. The MOVs are in series with 30A/700V fuses. Neon indicators are in parallel with the MOVs and are on unless the facility main breaker is open, or one or more MOV fuses are blown. These indicators operate as phase detectors.

The main input breaker CB1 turns power on and off to the PowerTECH regulator. The circuit breaker is shunt tripped by the bypass switch, the EPO (emergency power off) switch, or an SCR failure. When the bypass switch is rotated (Auto or Manual positions) the main input breaker is automatically tripped. The bypass switch is in the Auto position under normal operating conditions. The EPO switch trips the main input breaker when pressed and turns power off to the load. When an SCR malfunctions, the SCR fail light goes on and, if the bypass switch is in the Auto position, the main input breaker is tripped. An RC network across the input terminals attenuates voltage spikes that occur when the main input breaker is turned on.

The control logic board current detectors monitor each line current and compare each signal to a preset level in order to activate the regulation circuit. The voltage detector monitors each input line-to-line voltage from T2, T3, and T4. The signal is compared to a reference voltage level. The output of the voltage detector is fed to the comparator network. The comparator network selects the appropriate transformer tap. The output of the comparator network is fed to a voltage flip-flop. The voltage flip-flop transfers the input tap selection information to the output stage through the current flip-flop when a clock signal is received.

The clock signal comes from the change detector circuit. The change detector circuit monitors the input and output of the voltage flip-flop once per cycle. When the input of the voltage flip-flop is different from the output, the change detector circuit generates three outputs. One output turns off all SCR drive circuits through the output stage. The conducting SCR continues to conduct until there is zero current. The second output transfers the tap selection information from the input to the output on the voltage flip-flop. The third output provides a clock signal to the current flip-flop. The output stage on the control logic board acts as an interface to the SCR driver board. The master reset circuit on the control logic board inhibits the output for six seconds and is activated by the main input breaker, or the under voltage detection circuit. The undervoltage detection circuit compares the input voltage signal to a preset reference of 3V. When the voltage falls below the reference level, the SCR drive circuit is turned off. The undervoltage shutdown occurs when the line voltage sags by 40% for at least two cycles.

The SCR driver board has a 5 kHz oscillator that provides a push/pull signal to MOSFETs that energize the primary of the pulse transformer. The pulse transformer output is applied to the SCR bank through snubber boards.

There are six SCRs per phase mounted on a heat sink. The SCRs are fused together by copper bus wires (fuse links). At any time, only one SCR is conducting. If two or more SCRs begin to conduct at the same time, some fuse links will open.

The SCR failure sensing board monitors the fuse link continuity, and shunt trips the main input breaker, and provides a signal for the SCR Failure LED if this circuit is open. Additional functions of the SCR failure sensing board are: providing input, output, and neutral signals to the front panel test points; monitoring the transformer temperature, and providing a shunt trip signal to the main input breaker in an over temperature condition ($>160^{\circ}\text{C}$); and providing signals for the EPO and Reset switches.

The capacitors across the output line-to-neutral terminals smooth out any waveform distortion that may arise due to tap switching. Note that the neutral and ground buses are tied together. The 3-phase output is distributed within the 3-phase load center. The load center contains all feeder circuit breakers. From the load center, subsystem power is distributed through contactors to each subsystem. Feeder 17 (PDU Power) and Feeder 16 (Service Receptacles) are the only feeders which are not contactor-controlled.

3- MODES OF OPERATION

3-1 Introduction - Sheet 2 of 5

There are five operating modes defined for the PDU with specific groups of subsystems receiving power under each one. The PDU is responsible for sequencing this power to the groups in an orderly manner. The logical grouping of the subsystems is shown in Table 1 below. Refer to Table 2 under Table 1 for the modes of operation and their associated group status.

TABLE 1
SUBSYSTEM LOGICAL GROUPING

Note

Contactors K4, K6, K8, K11, and Circuit Breakers CB7, CB12, CB13, CB14, and CB16 (marked with asterisks (*) in the table) are not included with newer standard PDUs. Wiring to these contactors and circuit breakers is included, but wire ends are capped. K8 and CB16 may be included if system has resistive shim coil.

GROUP	COMPONENT	CONTACTOR	FEEDER	CIRCUIT BREAKER
A	RF Cabinet (MR1)	K10	1	CB17
	Gradient Amplifier Cabinet (MR3)	K9	4	CB6
	Penetration Cabinet (PP2)	K12	15	CB8
	Broadband RF Amplifier Cabinet (MR6) (Optional)	K11*	5	CB7*
B	Shim Power Supply Cabinet (MG4/MM4/MS3)	K8*	6	CB16*
	Main Magnet Power Supply Cabinet (MG5/MM5/MS2)	K7	7	CB15
C	Spare	K6*	3	CB12*
D	System Control Cabinet (MR2)/Computer Cabinet (CC1)	K3	10	CB11
E	Operator Console (OC1)	K1	9	CB1
	Independent Console (IC1) (Optional)	K5	2	CB5
	Spare	K2	12	CB4
	Spare	K2	11	CB3
	Multiformat Unit (MF1)/Laser Camera (LC1) (Optional)	K2	8	CB2
	Spare	K4*	13	CB13*
	Spare		14	CB14*
F	PDU Power (PD1)	N/A	17	CB10
	Service Receptacles	N/A	16	CB9

TABLE 2
PDU OP MODES AND GROUP

MODE	GROUP/STATUS					
	A	B	C	D	E	F
#1 Emergency Off	OFF	OFF	OFF	OFF	OFF	OFF
#2 Full Off	OFF	OFF	OFF	OFF	OFF	ON
#3 Emergency Stop	OFF	OFF	OFF	ON	ON	ON
#4 Overnight	OFF	ON	ON	ON	ON	ON
#5 Full On	ON	ON	ON	ON	ON	ON

3-2 Mode Selection - Sheet 1 or 4

Under normal operation, the PDU control board selects modes 2, 4, and 5. In addition, mode 2 can be forced by an over temperature condition in the system cabinet, and mode 4 can be forced by an over temperature condition in the quad/Roemer RF/gradient coil. Some upgraded systems have a five-button PDU remote control panel on the system cabinet that can also be used to select modes 2, 4, and 5. Mode 3 is normally selected with the Emergency Stop buttons on the operator work space, or the magnet enclosure, although it is selectable from the PDU control board, or the PDU remote control panel. The PDU control board is mounted on the PDU cabinet door and the PDU remote control panel is located in the system cabinet.

An interlock on the PDU cabinet door disables the PDU remote control panel when the door is open for service. An additional jack (J3) on the PDU control board is not disabled when the door is open. This jack is pin-for-pin compatible with the jack for the PDU remote control panel (J2 on the PDU front door) and serves the same functions. J3 is used if remote control of the PDU is required during servicing with the front door open.

3-3 Mode Descriptions

3-3-1 Mode #1 - Emergency Off

Nonregulated PDU - Sheet 1

This mode is actuated either by pushing one of the red mushroom buttons (customer-supplied) located at the exits, or when the temperature within the PDU exceeds the maximum limit (refer to Section 4-1 Over temperature Monitors). All primary power to the PDU is removed in this mode since the facility disconnect will be tripped. System power can be restored only after resetting the facility disconnect. This mode should not be confused with Emergency Stop (mode #3 below).

PDU with PowerTECH Regulator - Sheet 3

This mode is actuated either by pushing one of the customer supplied Emergency Off buttons or when the regulator shunt trips the input breaker (CB1) (Refer to Section 4-1 Over temperature Monitors). Incoming power to the PDU is removed when the Emergency Off button is actuated since the facility disconnect is tripped. Power is still present on the input side of the input breaker when CB1 is tripped by the over temperature shunt. This mode should not be confused with Emergency Stop (mode #3 below).

3-3-2 Mode #2 - Full Off - Sheet 1 or 4

All subsystems are off while in Full Off mode, however, PDU power (Feeder 17) is on and all service outlets (Feeder 16) are powered on. This mode is automatically enabled after initially applying system power to the PDU through the facility disconnect. The PDU is also placed in this mode when the system cabinet is in an over temperature condition (refer to Section 4-1 Over temperature Monitors).

3-3-3 Mode #3 - Emergency Stop - Sheet 1 or 4

Emergency Stop is used only under certain emergency situations related to the patient vicinity. When any Emergency Stop button is pushed, power to most components within the patient vicinity is removed. Power to the system cabinet, computer cabinet, computer peripherals, and operator work space is not removed.

The Emergency Stop circuit is a normally closed loop that originates on the PDU control board. Relay coil K41 is normally energized through the loop by the PDU +24Vdc power supply. If the loop is opened by pushing an Emergency Stop button on the operator work space, or at the magnet enclosure, K41 drops out. This provides a return for K16 coil through the K15, K41, and K26 contacts. (The K15 contacts are closed since Full Off mode is not selected. As expected, you cannot go from Full Off mode to Emergency Stop mode because of K15. K26 is a time delay relay which is normally energized as long as power to the PDU is applied.) K13 (not on the PDU Control Board) now drops out and contactors K6 through K12 are also forced to drop out (groups A, B, and C in Table 1 above in Section 3-1).

When Emergency Stop mode is selected directly at the PDU control board, or the PDU remote control panel in the system cabinet, K16 is energized through the K15 normally closed contacts. As before, K13 drops out causing contactors K6 through K12 to drop out.

The Emergency Stop loop is fed to the SCB via J2 on the PDU control board (signals EMSTOP and LVLERETURN). Once in the SCB, it is routed to the operator work space, and the magnet enclosure. Note that this loop is not buffered anywhere in the MR system.

The SCB also monitors the Emergency Stop push button status from both the operator work space (C-EMERG), and the magnet enclosure (EMER-STOP). When a switch closure is detected, a message indicating an emergency stop occurred will be displayed on the operator work space.

3-3-4 Mode #4 - Overnight - Sheet 1 or 4

In the overnight mode, all subsystems are powered on except for major energy consumers (e.g., power amplifiers) and subsystems that may exhibit mechanical wear when idling. This mode is used to conserve power and reduce wear between normal operating hours. This mode is normally selected while running overnight diagnostics (although it is not required). For systems with quad/Roemer RF/gradient coils, this mode can also be enabled by an over temperature condition in the gradient coil (refer to Section 4-1 Over temperature Monitors).

3-3-5 Mode #5 - Full On - Sheet 1 or 4

This is the mode used for normal operation (i.e., imaging). All subsystems are powered on.

3-4 Mode Sequencing - Sheet 2 or 5

The PDU sequences power in a controlled fashion during mode transitions. Table 3 below describes the sequences for each transition. Only those transitions described in Table 3 are allowed by the PDU.

TABLE 3
PDU POWER SEQUENCING

TRANSITION	SEQUENCE (Refer to Table 1 for subsystem groups)
Emerg. Off → Full Off	All Off, F
Full Off → Emerg. Off	F, All Off
Full Off → Overnight	F, EF, Wait T1, DEF, CDEF, wait T2, BCDEF
Full Off → Full On	F, EF, Wait T1, DEF, CDEF, wait T2, BCDEF, ABCDEF
Emerg. Stop → Emerg. Off	DEF, All Off
Emerg. Stop → Full Off	DEF, EF, wait T4, F
Emerg. Stop → Overnight	DEF, CDEF, wait T2, BCDEF
Emerg. Stop → Full On	DEF, CDEF, wait T2, BCDEF, ABCDEF
Overnight → Emerg. Off	BCDEF, All Off
Overnight → Full Off	BCDEF, CDEF, wait T3, DEF, EF, wait T4, F
Overnight → Emerg. Stop	BCDEF, DEF
Overnight → Full On	BCDEF, ABCDEF
Full On → Emerg. Off	ABCDEF, All Off
Full On → Full Off	ABCDEF, BCDEF, CDEF, wait T3, DEF, EF, wait T4, F
Full On → Emerg. Stop	ABCDEF, DEF
Full On → Overnight	ABCDEF, BCDEF
Note: T1 = Time for computer to stabilize. T2 = Time for micro-based systems to stabilize. T3 = Time for energy stored in groups A and B to decay T4 = Time necessary to prevent spurious signals to audible or visual indicators. T1 = T2 = T3 = T4 = 5 seconds.	

4- FAULT / MODE MONITORING

4-1 Overtemperature Monitors - Sheet 1, 3, or 4

The PDU power transformer (nonregulated PDUs), the PowerTECH Regulator (regulated PDUs), the system cabinet, and the quad/Roemer RF/ gradient coil are monitored for over temperature conditions. The methods of handling these conditions differ as described below.

PDU Overtemperature Monitoring (Nonregulated PDUs)

Two sets of temperature sensors are mounted within the power transformer. Each set contains three sensors in parallel. The sensors of one set trip when the transformer temperature exceeds 329°F (165°C). This condition enables the signal TEMPCAUTION which energizes relay K40. One K40 contact closes causing the OVERTEMP indicators to turn on. The other contact opens and alerts the SCB to the condition through the TEMPALERT signal.

The second set of temperature sensors trip when the transformer exceeds 392°F (200°C). This enables the signal TEMPEXTREME, energizing relay K42 on the PDU control board. In addition to tripping the facility main circuit breaker, signal SETIND is pulsed through a K42 contact to an electromechanical indicator within the PDU. This temperature exceeded indicator retains its indication of the fault state after the loss of primary power. When primary power is restored, pressing either the Full Off, Overnight or Full On push buttons resets the indicator by enabling the CLEARIND signal. Both CLEARIND and SETIND are derived from the +12Vdc regulator circuit on the PDU control board.

4-1-1 PDU with PowerTECH Regulator Overtemperature Monitoring

A sensor in the regulator sends a shunt trip signal to the input breaker when the regulator transformer temperature exceeds 320°F (160°C). Power is still present on the input side of the input breaker when that breaker is tripped by the over temperature shunt.

4-1-2 System Cabinet Overtemperature Monitoring

The system cabinet (MR2) contains a normally open sensor that monitors the temperature within the cabinet. The sensor trips at 105°F (40.6°C), sending a signal to the Temperature Sense Control Box located within the system cabinet. K1 in the Temperature Sense Control Box (MR2A18) is a time delay relay. When the sensor trips, K1 is energized after a delay of about 10 minutes. While K1 is counting down, an audible alarm (DS2) sounds. When K1 finally is energized, its contacts close causing the PDU to enter the Full Off mode.

4-1-3 Quad/Roemer RF/Gradient Coil Overtemperature Monitoring

Two temperature sensors are contained within the quad/Roemer RF/gradient coil. One sensor trips when the gradient coil temperature exceeds 212°F (100°C). This enables the signal 100C SENSOR which energizes relay K21. The K21 contact opens and alerts the SCB to the condition through the 100C OVERTEMP signal.

The second temperature sensor trips when the gradient coil temperature exceeds 248°F (120°C). This enables the signal NIGHT, which initiates Overnight mode in the same fashion as if it had been initiated from the PDU control board, or the PDU remote control panel (on systems that have a five-button PDU remote control panel).

4-2 Mode Sense - Sheet 1

The SCB monitors the following PDU modes of operation: Emergency Stop, Overnight, Full On, and Over temperature. The associated signals are EMSTOP, STOPMODE, NIGHTMODE, and ONMODE respectively. These signals are pulled low when the mode relay coils on the PDU control board are energized (K16, K17, K18 respectively). Along with TEMPALERT, 100°C OVERTEMP, and RETURN (which is tied to +24Vdc), the mode sense signal lines go to the SCB CPU where all but RETURN are transformed to logic levels and sent to the TYME Board in the TPS. These signals are processed through the IPG Board and sent by the TPS to the operator work space (via the VME I/F boards) for display whenever a switch is activated.

Visual indication of the modes of operation and the over temperature indicator are given on the PDU control board and the PDU remote control panel located in the system cabinet. The five-button PDU remote control panel indicates all modes and over temperature, the two-button PDU remote control panel indicates only Emergency Stop and over temperature.

4-3 Elapsed Time Indicators - Sheet 1 or 4

Four elapsed time indicators within some models of the PDU monitor the Full Off, Emergency Stop, Overnight, and Full On modes of operation. These indicators operate off the 24Vdc power supply to ensure accuracy for both 50Hz and 60Hz systems. All are nonresettable. Each has a six-digit readout with a range of 0–99999.9 hours.

Note that all four indicators run when in the Full On mode. In overnight mode, all but the Full On indicator are active. Both the Full Off and Emergency Stop indicators run in the Emergency Stop mode. Only the Full Off indicator runs when in that respective mode.

Note

Contactors K4, K6, K8, K11, and Circuit Breakers CB7, CB12, CB13, CB14, and CB16 (marked with asterisks (*) in the table) are not included with newer standard PDUs. Wiring to these contactors and circuit breakers is included, but wire ends are capped. K8 and CB16 may be included if system has resistive shim coil.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 16, 1998	M. Whitlow	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.